

Sex determination of three Neotropical canids by high resolution melting analysis

Susana González^{1,2} · Natalia Mannise¹ · Leticia Repetto¹ · Jesús E. Maldonado^{3,4}

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Abstract We describe a fast and reliable genetic method for sexing Neotropical canids by high resolution melting analysis of amplicons obtained by real time polymerase chain reaction. We used a primer set that amplifies a 195 bp fragment of the *ZFX* and *ZFY* genes to determine the sex of pampas fox (*Pseudalopex gymnocercus*), crab eating fox (*Cerdocyon thous*), maned wolf (*Chrysocyon brachyurus*), and domestic dog (*Canis familiaris*). This new method is useful, fast and reliable for sexing of non-invasively obtained samples with low DNA quality and quantity. This protocol could be easily modified to sex non-invasive samples of other canid species.

Keywords *ZFY–ZFX* · Crab eating fox · Pampas fox · Maned wolf

We designed a method to rapidly and efficiently determine the sex of three species of wide-ranging South American

canids: the crab eating fox, (*Cerdocyon thous*), the pampas fox, (*Pseudalopex gymnocercus*), and the threatened maned wolf, (*Chrysocyon brachyurus*) and the domestic dog (*Canis familiaris*). Different sample sources were tested such as tissue recovered from carcasses, hair, blood, saliva and feces (Table 1). We used canid-specific sexing primers previously described by Ortega et al. (2004) to perform PCR on these samples, and analyzed the dissociation curve patterns generated by a High Resolution Melting channel on the Rotor Gene 6000[®] (Corbett Research). The differences in the dissociation curve patterns are due to differences in the base composition of the amplicons (Ririe et al. 1997; Palais et al. 2005; Montgomery et al. 2007; Taylor 2009; Smith et al. 2010; Henri and Mouton 2012; Morinha et al. 2013).

We extracted DNA from a total of 93 samples of known species and sex, collected over a 13-year period (1997–2013) and stored at the Conservation Genetics Laboratory at *Departamento de Biodiversidad y Genética-IIBCE-Uruguay* (Table 1). We also included, as experimental controls, a total of 37 fecal DNA that had been previously identified to species by sequencing short fragments of the mtDNA *D-loop* and/or the *Cyt b* regions (González 2005; Mannise et al. 2012), and sexed using the PCR–RFLP method described by Ortega et al. (2004).

DNA extraction from tissue samples were performed following the procedures described in Medrano et al. (1990) with optimizations performed as in González (1997), and fecal samples using the QIAGEN[®] DNeasy blood and tissue extraction kit with minor modifications to the manufacturer's protocol. We used *ZFKF203L* and *ZFKF195H* primers previously described by Ortega et al. (2004) to amplify a short 195 bp fragment of the *ZFX* and *ZFY* genes. PCR amplifications were performed in a final volume of 25 µl following the instructions of the *Immomix*[™] (Bioline) and adding 1 µl of *Syto 9* as fluorochrome. Reactions contained

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✉ Susana González
sgonzalez@iibce.edu.uy

¹ Departamento de Biodiversidad y Genética, Instituto de Investigaciones Biológicas Clemente Estable, Av. Italia 3318, 11600 Montevideo, Uruguay

² Sección Genética Evolutiva, Facultad de Ciencias UdelaR, Montevideo, Uruguay

³ Center for Conservation and Evolutionary Genetics, Smithsonian Conservation Biology Institute, National Zoological Park, Washington, DC, USA

⁴ Department of Vertebrate Zoology, National Museum of Natural History, Smithsonian Institution, Washington, DC, USA

Table 1 Average melting peak values in degrees centigrade of the *ZFX* and *ZFY* fragments obtained for each canid species from different sources of samples (tissue, hair, blood and fecal) and the corresponding calculated variance values obtained by Rotor Gene 6000® (Corbett Research)

Species	Samples	N	Male	Female	<i>ZFY</i> (°C)	S ²	<i>ZFX</i> (°C)	S ²
<i>Cerdocyon thous</i>	Tissue	26	8	18	80.696	0.101	81.553	0.03
	Hair	1		1	0	0	81.58	0
	Blood	4	4	0	80.54	0	81.7	0
	Feces	33	6	27	80.609	0.091	81.495	0.114
	Total	64	18	46	80.615	0.06	81.582	0.04
<i>Pseudalopex gymnocercus</i>	Tissue	7	2	5	80.65	0.007	81.668	0.05
	Hair	1		1	0	0	81.95	0
	Blood	4	2	2	80.4	0	81.67375	
	Feces	3		3			81.9	0
	Total	15	4	11	80.525	0.007	81.831	0.05
<i>Chrysocyon brachyurus</i>	Tissue	4	3	1	80.45	0.06	81.708	0.335
	Hair	2	1	1	80.45	0	81.77	0
	Blood	3	1	2	80.46	0	82.1	0
	Saliva	1		1			82.43	0
	Feces	1	1	0	80.46	0	81.49	0
<i>Canis familiaris</i>	Total	11	6	5	80.463	0.06	81.93	0.335
	Blood	3	2	1	80.35	0	81.385	0
Total	Total	3	2	1	80.35	0	81.385	0
	Total	93	30	63				

an initial concentration of 10 µM of the forward primer *ZFKF203L* and 10 µM of the reverse primer *ZFKF195H*, and 30 ng of DNA template. We developed a protocol with three different profiles. We included an initial denaturation step of 95 °C for 10 m and the following cycles: (1) of 95 °C for 20 s, 60 °C for 20 s and 72 °C for 20 s. repeated for 10 times, (2) of 95 °C for 20 s, 59 °C for 20 s and 72 °C for 20 s. repeated for 15 times, and (3) of 95 °C for 20 s, 58 °C for 20 s and 72 °C for 20 s. repeated for 20–25 times. Finally an extension step of 72 °C for 15 m. The HRM step included a 65–95 °C ramp increasing at 0.1 °C per step with a 2 s pause at each step.

We sequenced two individuals per each species for fragments derived from the *ZFY* and *ZFX* genes. We retrieved domestic dog sequences from GenBank for alignment and analysis of sequence variation (*ZFX* AF027765.1 and *ZFY* AF027766.1). We determined that females from all species were identical and had a unique restriction site for *HaeIII*. Since both loci amplify PCR products of the same size (195 bp), we isolated the *ZFY* fragment by conducting a locus specific enzymatic digestion of the PCR products from known males in a volume of 20 µl which contained 10 µl of PCR product, ddH₂O, 2 µl of NEB buffer (10×) *HaeIII* and 5 U of restriction enzyme *HaeIII* (New England Biolabs). Products were run slowly (<40 V) in 3 % agarose gels stained with GoodViewTM, yielded one Y band of 195 bp and two bands for the X amplicon one of 157 bp and 38 bp. The *ZFY* fragment was gel extracted using a gel extraction ZymocleanTM Gel DNA Recovery Kit (Zymo Research).

We conducted an analysis of variance (ANOVA) to test if the melting peaks were significantly different in the *ZFY*-*ZFX* fragments among species, and from different DNA source types (tissue or feces) (StatSoft 1995).

The protocol designed unambiguously assigns the sex of the 93 control samples and was 96 % effective for fecal DNA (Table 1). As expected, the HRMA, showed a pattern of two melting curves in the case of known sex males and a single curve in the case of known sex females in all of the samples tested (Fig. 1).

The high resolution melting curves for the *ZFX* of all four species and individuals were identical. The ANOVA results for *ZFX* melting peaks temperatures were not significant among species ($F = 0.14$; $p = 0.737$). As expected, the *ZFX* sequenced fragments did not reveal any mutational differences for the four species.

The *ZFY* melting curves were different among species and the ANOVA results showed a significant F value for the *ZFY* variance ($F = 169$; $p = 0.005$). The pampas fox had the same *ZFY* melting curve pattern as the domestic dog, but the melting curve profiles (Fig. 1) are influenced by sequence variations in the crab eating fox (at position 8 G/C; JQ946321) and the maned wolf (at position 5 T/C; JQ946322). Furthermore we performed an ANOVA to analyze a relatively large number of crab eating fox samples and determined that the DNA sample source type did not significantly affect the melting temperature peaks of the *ZFY* and *ZFX* amplicons.

The advantage of this method is fast in less than 2.5 hours is possible to unambiguously determine sex from

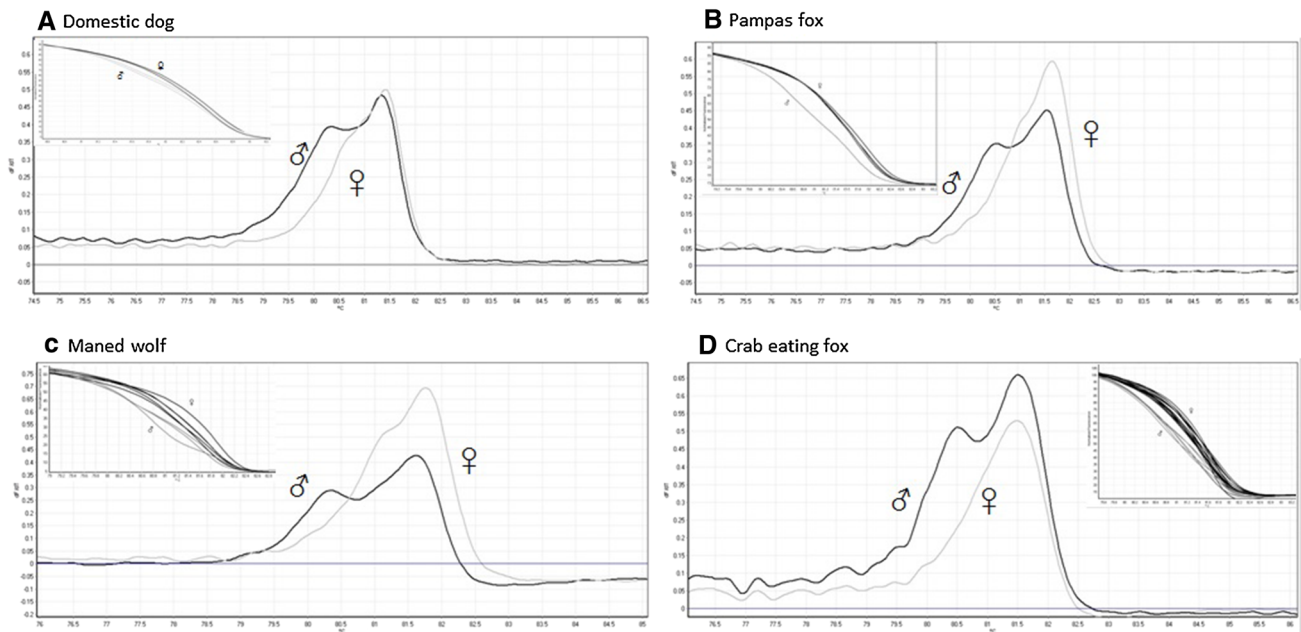


Fig. 1 The high resolution melting curves for each species showing the ZFX (one single peak corresponding to female) and ZFY (two peaks corresponding to male) amplicon patterns and in the inset is the normalized graph that allows to differentiate males from females.

different DNA sources and quality. Additionally substantial saving of laboratory consumables (almost five times cheaper), and also reducing contamination risk. Moreover, HRMA is a novel and useful tool for performing population genetic surveys using noninvasive sampling methods that focus on fecal DNA for sex determination of native canids (Ralls et al. 2010).

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- a Domestic dog (*Canis familiaris*), b pampas fox (*Pseudalopex gymnocercus*), c maned wolf (*Chrysocyon brachyurus*) and d crab eating fox (*Cerdocyon thous*)

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